Crop Recommendation System Using Machine Learning

A Project Report Submitted in the
Partial Fulfillment of the Requirements
for the Award of the Degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

Aknapuram Koushik Reddy	**23881A0403**
Almala Chaitanya	**23881A0405**
Doma Meghana	**23881A0411**

SUPERVISOR
Dr K Ashwini
Assistant Professor,ECE

VARDHAMAN
COLLEGE OF ENGINEERING

Department of Electronics and Communication Engineering

June, 2026

VARDHAMAN
COLLEGE OF ENGINEERING

Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the project titled **Crop Recommendation System Using Machine Learning** is carried out by

Aknapuram Koushik Reddy	**23881A0403**
Almala Chaitanya	**23881A0405**
Doma Meghana	**23881A0411**

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Electronics and Communication Engineering** during the year 2025-26.

Signature of the Supervisor
Dr K Ashwini
Assistant Professor,ECE

Signature of the HOD
Prof.S.Rajendar
Professor and Head, ECE

Acknowledgements

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We wish to express our deep sense of gratitude to **Dr K Ashwini**, Assistant Professor,ECE Project Supervisor, Department of Electronics and Communication Engineering, Vardhaman College of Engineering, for his able guidance and useful suggestions, which helped us in completing the project in time.

We are particularly thankful to **Prof.S.Rajendar**, the Head of the Department, Department of Electronics and Communication Engineering, his guidance, intense support and encouragement, which helped us to mould our project into a successful one.

We show gratitude to our honorable Principal **Prof.J.V.R. Ravindra**, for providing all facilities and support.

We avail this opportunity to express our deep sense of gratitude and heartful thanks to **Dr. Teegala Vijender Reddy**, Chairman and **Sri Teegala Upender Reddy**, Secretary of VCE, for providing a congenial atmosphere to complete this project successfully.

We also thank all the staff members of Electronics and Communication Engineering department for their valuable support and generous advice. Finally thanks to all our friends and family members for their continuous support and enthusiastic help.

Aknapuram Koushik Reddy

Almala Chaitanya

Doma Meghana

Abstract

Agriculture is one of the most important sectors contributing to food security and economic development. Selecting the right crop based on soil nutrients and environmental conditions is essential for improving crop yield and reducing agricultural losses. This project, Crop Recommendation System Using Machine Learning, is designed to assist farmers in making informed decisions by recommending the most suitable crop for cultivation based on various agricultural parameters.

The system utilizes a Random Forest Classifier, a supervised machine learning algorithm known for its high accuracy and reliability in classification tasks. The model is trained using a crop recommendation dataset containing features such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall. Based on these input values, the system predicts the most appropriate crop for cultivation.

Apart from recommending a crop, the application provides several additional features to improve decision-making. These include prediction confidence, fertilizer suggestions, soil status analysis (Acidic, Neutral, or Alkaline), water requirement, rainfall status, suitability rating, top three recommended crops, crop information, and farmer advice. These features make the system more practical and informative for agricultural applications.

The project is implemented using Python and developed with libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib for data preprocessing, model training, evaluation, and visualization. The Random Forest model achieved an accuracy of 99.32%, demonstrating its effectiveness in predicting suitable crops from the given dataset.

This project aims to promote smart agriculture by combining machine learning with agricultural knowledge. In the future, the system can be enhanced by integrating real-time weather APIs, deploying it as a Flask or Streamlit web application, or developing a mobile application to provide farmers with real-time crop recommendations and advisory services.

Keywords: Machine Learning, Crop Recommendation, Random Forest, Smart Agriculture, Soil Analysis, Crop Prediction, Python, Scikit-learn.

Table of Contents

List of Tables

List of Figures

CHAPTER 1

Introduction

1.1 Introduction

Agriculture plays a significant role in the economy and is the primary source of livelihood for millions of people around the world. The success of agriculture largely depends on selecting the right crop according to soil fertility, climatic conditions, and environmental factors. Traditionally, farmers rely on personal experience, seasonal knowledge, or expert advice to choose suitable crops. However, inaccurate crop selection can result in poor yield, financial losses, and inefficient utilization of available resources such as water, fertilizers, and land.

With the advancement of Machine Learning (ML), intelligent decision-support systems can analyze agricultural data and provide accurate crop recommendations. Machine learning algorithms identify patterns from historical datasets and predict the most suitable crop based on different soil and environmental parameters. This enables farmers to make data-driven decisions, leading to increased productivity and sustainable farming practices.

The Crop Recommendation System Using Machine Learning is developed to recommend the most appropriate crop using important agricultural parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall. The project employs the Random Forest Classifier, a robust ensemble machine learning algorithm known for its high prediction accuracy and reliability. The trained model analyzes the input values and predicts the crop that is most suitable for cultivation.

In addition to crop prediction, the proposed system provides several useful features including prediction confidence, fertilizer suggestion, soil status analysis, water requirement, rainfall status, suitability rating, top three recommended crops, crop information, and farmer advice. These features assist farmers in making better agricultural decisions and improving crop management practices.

The project is implemented using Python with libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib for data processing, model training, evaluation, and visualization. The developed model achieved an accuracy of 99.32 , demonstrating its effectiveness in recommending crops based on the given agricultural conditions.

This project aims to promote smart agriculture by integrating machine learning techniques with agricultural knowledge. It helps farmers improve crop productivity, optimize resource utilization, and reduce the risk of unsuitable crop selection. In the future, the system can be extended by integrating real-time weather APIs, deploying it as a web application using Flask or Streamlit, or developing a mobile application to provide farmers with real-time crop recommendations and advisory services.

1.2 Background and Motivation

Agriculture is the backbone of many economies and plays a crucial role in ensuring food security and sustainable development. Farmers often face challenges in selecting the most suitable crop due to changing climatic conditions, varying soil fertility, and limited access to scientific decision-making tools. Traditional crop selection methods are generally based on personal experience or local practices, which may not always provide the best results. As a result, farmers may experience low crop yield, excessive use of fertilizers, increased production costs, and financial losses.

The rapid growth of Machine Learning (ML) has created new opportunities to improve agricultural practices through intelligent decision-support systems. By analyzing historical agricultural data, machine learning models can identify relationships between soil nutrients, environmental conditions, and crop suitability, enabling accurate crop recommendations. Such systems help farmers make informed decisions, optimize resource utilization, and improve agricultural productivity.

The motivation behind this project is to develop a Crop Recommendation System Using Machine Learning that assists farmers in selecting the most suitable crop based on important soil and environmental parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall. In addition to crop prediction, the system provides prediction confidence, fertilizer suggestion, soil status, water requirement, rainfall status, suitability rating, crop information, and farmer advice, making it a practical decision-support tool.

By integrating machine learning with agriculture, this project aims to promote smart farming, reduce crop failure, improve productivity, and encourage efficient utilization of agricultural resources. The proposed system demonstrates how data-driven technologies can support sustainable farming and enhance decision-making for farmers.

1.2.1 Objectives of the Project

The main objective of this project is to develop a Machine Learning-based Crop Recommendation System that assists farmers in selecting the most suitable crop based on soil and environmental conditions. The project aims to improve agricultural decision-making by providing accurate crop recommendations and additional advisory information.

- To develop a crop recommendation system using the Random Forest Classifier.
- To recommend the most suitable crop based on Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall.
- To achieve high prediction accuracy for reliable crop recommendations.
- To provide additional features such as prediction confidence, fertilizer suggestion, soil status, water requirement, rainfall status, suitability rating, top three crop recommendations, crop information, and farmer advice.
- To assist farmers in making informed decisions and improving crop productivity.
- To promote smart and sustainable agriculture using machine learning techniques.

- To provide a foundation for future enhancements such as real-time weather integration and web or mobile application deployment.

1.2.2 Organization of the Project

This project is organized into different phases to ensure systematic development and implementation of the Crop Recommendation System.

- **Chapter 1:** Introduction: Presents the background, motivation, objectives, and significance of the project.
- **Chapter 2:** Literature Survey: Reviews existing research and machine learning approaches used for crop recommendation.
- **Chapter 3 :**System Analysis and Design: Describes the problem statement, system architecture, dataset, and design methodology.
- **Chapter 4:** Methodology: Explains data preprocessing, feature selection, model training using the Random Forest Classifier, and prediction process.
- **Chapter 5:** Implementation and Results: Presents the implementation details, model accuracy, output screenshots, and additional features such as fertilizer suggestion, soil status, water requirement, prediction confidence, top three crop recommendations, and farmer advice.
- **Chapter 6:** Conclusion and Future Scope: Summarizes the project outcomes and discusses possible future enhancements, including weather API integration and web or mobile application deployment.

CHAPTER 2

Literature Survey

2.1 Introduction

The literature survey provides an overview of the existing research and technologies related to crop recommendation systems. It helps in understanding the methods, algorithms, datasets, and techniques used by researchers to improve agricultural productivity through machine learning. Several studies have focused on predicting suitable crops based on soil nutrients, climatic conditions, and environmental factors to support farmers in making informed decisions.

Traditional crop recommendation methods mainly depend on farmers' experience and manual analysis, which may not always produce accurate results due to changing environmental conditions. With the advancement of Machine Learning (ML), intelligent systems have been developed to analyze agricultural data and recommend suitable crops with higher accuracy. Various machine learning algorithms such as Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naïve Bayes, and Random Forest have been applied for crop prediction, each offering different levels of performance.

After reviewing the existing literature, the Random Forest Classifier was selected for this project because of its high prediction accuracy, robustness, and ability to handle multiple input features effectively. The proposed system uses important agricultural parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend the most suitable crop. In addition, the project includes useful features such as fertilizer suggestion, soil status analysis, water requirement, prediction confidence, top three crop recommendations, and farmer advice, making it more practical and beneficial for farmers.

2.2 Identified Research Gaps

Based on the literature survey, several limitations were identified in existing crop recommendation systems. Many existing models focus only on predicting the most suitable crop and do not provide additional guidance that can help farmers make better agricultural decisions. Most systems also consider a limited set of features and lack practical advisory information.

The following research gaps were identified:

- Existing systems mainly provide crop prediction without additional recommendations.
- Many models do not include fertilizer suggestions based on the recommended crop.
- Limited information is provided about soil status, water requirement, and rainfall conditions.
- Most systems recommend only a single crop instead of providing alternative crop options.
- Few crop recommendation systems provide a prediction confidence score, making it difficult for users to assess the reliability of the prediction.
- Existing solutions often lack farmer advisory information, which can help improve crop cultivation practices.
- Most models are designed as research prototypes and are not easily accessible through user-friendly applications.

To address these gaps, the proposed Crop Recommendation System Using Machine Learning not only predicts the most suitable crop but also provides prediction confidence, fertilizer suggestion, soil status analysis, water requirement, rainfall status, suitability rating, top three crop recommendations, crop information, and farmer advice, making the system more practical and useful for farmers.

2.3 Summary

This chapter presented a review of existing research related to crop recommendation systems and the application of machine learning in agriculture. Various machine learning algorithms, including Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naïve Bayes, and Random Forest, were studied to understand their effectiveness in crop prediction. The literature survey also highlighted the advantages and limitations of existing systems.

Based on the analysis, several research gaps were identified, including the lack of additional advisory features such as fertilizer recommendation, soil status analysis, water requirement, prediction confidence, and farmer guidance. To address these limitations, the proposed Crop Recommendation System Using Machine Learning utilizes the Random Forest Classifier to recommend the most suitable crop based on soil and environmental parameters. The system also incorporates additional features to provide comprehensive support for farmers.

CHAPTER 3

Review of Prior Research

3.1 Introduction

The review of prior research provides an understanding of the existing studies and developments in the field of crop recommendation using machine learning. Researchers have proposed various intelligent systems to improve agricultural productivity by analyzing soil properties, climatic conditions, and environmental factors. These systems assist farmers in selecting suitable crops, thereby improving crop yield and reducing agricultural losses.

Several machine learning algorithms, such as Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naïve Bayes, and Random Forest, have been applied in crop recommendation systems. These algorithms are evaluated based on prediction accuracy, computational efficiency, and their ability to handle agricultural data. Many studies have reported that the Random Forest Classifier provides better performance due to its robustness, high accuracy, and ability to manage complex datasets with multiple input features.

This section reviews the methodologies, datasets, algorithms, and findings of previous research works. The review helps in identifying the strengths and limitations of existing approaches and provides the foundation for developing the proposed Crop Recommendation System Using Machine Learning with additional features such as prediction confidence, fertilizer suggestion, soil status analysis, water requirement, top three crop recommendations, and farmer advice.

3.2 Hardware Implementation Considerations

The proposed Crop Recommendation System Using Machine Learning is a software-based application and does not require any specialized hardware components for its implementation. The system is developed and executed on a standard personal computer using Python and machine learning libraries. The hardware requirements include a computer with at least an Intel Core i3 processor or equivalent, 4 GB RAM (8 GB recommended), and sufficient storage space to accommodate the project files and dataset. The application is compatible with operating systems such as Windows, Linux, and macOS. Since the dataset is relatively small and the Random Forest Classifier is computationally efficient, the system can perform training and prediction with minimal hardware resources. These requirements ensure smooth execution of the application, making it accessible to students, researchers, and users without the need for expensive or specialized hardware.

3.3 Summary

This chapter presented the system design and implementation considerations for the proposed Crop Recommendation System Using Machine Learning. It described the overall architecture, methodology, dataset, software tools, and hardware requirements used for developing the system. The project utilizes the Random Forest Classifier to recommend the most suitable crop based on soil nutrients and environmental parameters. The chapter also discussed the software and hardware requirements necessary for efficient execution of the application. The proposed system is designed to provide accurate crop recommendations along with additional features such as fertilizer suggestion, soil status analysis, water requirement, prediction confidence, top three crop recommendations, and farmer advice. These design considerations ensure that the system is reliable, user-friendly, and suitable for supporting smart agricultural decision-making.

CHAPTER 4

Proposed Architecture

4.1 Introduction

The proposed architecture of the Crop Recommendation System Using Machine Learning is designed to recommend the most suitable crop based on soil and environmental parameters. The system follows a structured workflow, starting from data collection and pre processing to model training, prediction, and result generation. The input parameters include Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall, which are processed using the Random Forest Classifier. The trained model analyzes these inputs and predicts the most appropriate crop for cultivation. In addition to crop recommendation, the system provides prediction confidence, fertilizer suggestion, soil status, water requirement, rainfall status, suitability rating, top three crop recommendations, crop information, and farmer advice. This architecture ensures efficient data processing, accurate prediction, and a user-friendly output, making the system suitable for supporting smart agricultural decision-making.

4.2 Summary

This chapter presented the proposed architecture and methodology of the Crop Recommendation System Using Machine Learning. It explained the overall workflow of the system, including data collection, preprocessing, model training, prediction, and result generation. The Random Forest Classifier was used to analyze soil nutrients and environmental parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend the most suitable crop. The chapter also described the additional features implemented in the system, including prediction confidence, fertilizer

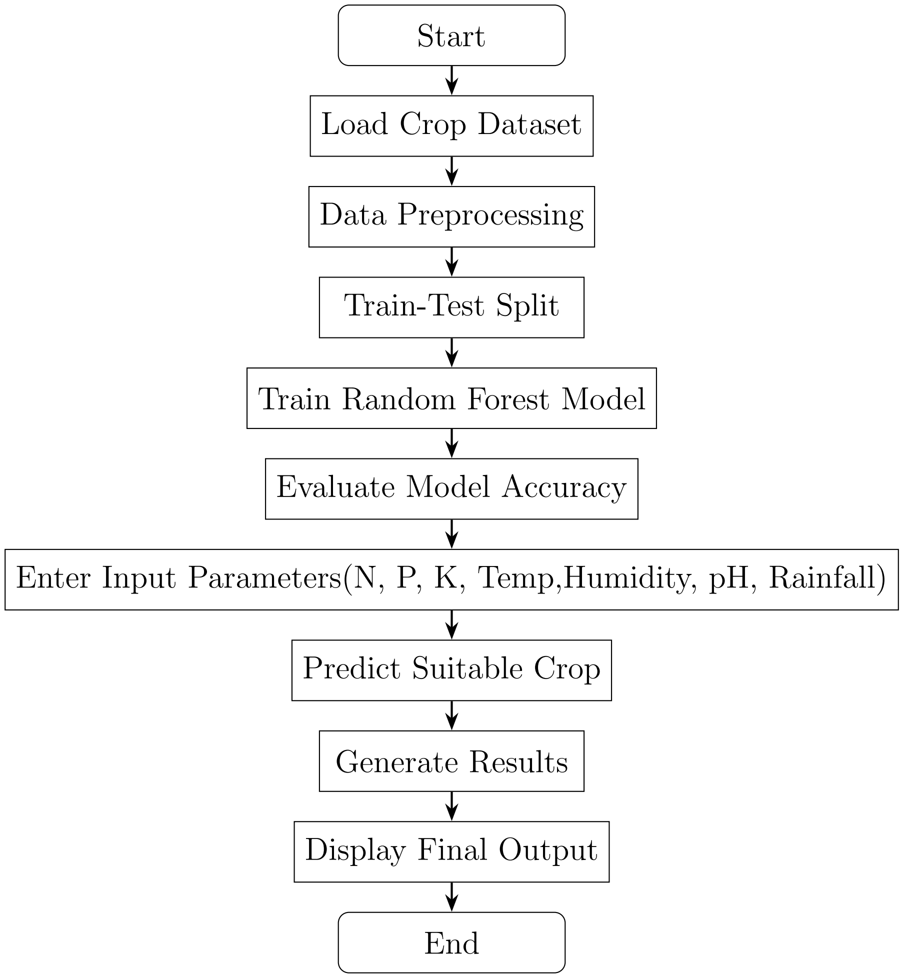

Figure 4.1: Flowchart of the Proposed Crop Recommendation System

suggestion, soil status analysis, water requirement, rainfall status, suitability rating, top three crop recommendations, crop information, and farmer advice. The proposed architecture ensures accurate crop prediction and provides useful agricultural recommendations, making the system reliable, efficient, and suitable for smart farming applications.

CHAPTER 5

Results and Discussions

5.1 Introduction

The Results and Discussion chapter presents the implementation outcomes and performance analysis of the proposed Crop Recommendation System Using Machine Learning. The system was developed using Python and implemented with the Random Forest Classifier to recommend the most suitable crop based on soil nutrients and environmental parameters. The model was trained and tested using the crop recommendation dataset containing features such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall.

The performance of the model was evaluated using prediction accuracy, and the Random Forest Classifier achieved an accuracy of 99.32 , demonstrating its effectiveness in crop prediction. The chapter also presents the system outputs, including the recommended crop, prediction confidence, fertilizer suggestion, soil status, water requirement, rainfall status, suitability rating, top three recommended crops, crop information, and farmer advice. The obtained results are analyzed and discussed to demonstrate the accuracy, reliability, and practical usefulness of the proposed system. These findings indicate that the developed system can serve as an effective decision-support tool for promoting smart and sustainable agricultural practices.

5.2 Simulation Environment

The proposed **Crop Recommendation System Using Machine Learning** was developed and tested in a Python-based simulation environment. The implementation was carried out using **Visual Studio Code (VS Code)** as the integrated development environment (IDE). The system utilizes the **Random Forest Classifier** from the **Scikit-learn** library for training and predicting the most suitable crop based on agricultural parameters. The crop recommendation dataset, containing features such as **Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall**, was used for model training and evaluation. Data preprocessing and manipulation were performed using the **Pandas** and **NumPy** libraries, while **Matplotlib** was used for data visualization and performance analysis. The developed model achieved an accuracy of **99.32%**, demonstrating its effectiveness in crop prediction. The simulation environment provides a reliable platform for testing the system and generating outputs such as crop recommendation, prediction confidence, fertilizer suggestion, soil status, water requirement, rainfall status, suitability rating, top three recommended crops, crop information, and farmer advice.

Table 5.1: Simulation Environment Specifications

Component	**Specification**
Programming Language	Python 3.x
IDE	Visual Studio Code (VS Code)
Machine Learning Algorithm	Random Forest Classifier
Libraries Used	Pandas, NumPy, Scikit-learn, Matplotlib
Dataset	Crop Recommendation Dataset (CSV)
Operating System	Windows 10/11
Model Accuracy	99.32%

5.3 Functional Verification

The functional verification of the proposed **Crop Recommendation System Using Machine Learning** was carried out to ensure that all system components operate correctly and produce the expected results. The system was tested using different combinations of soil nutrient values and environmental parameters, including **Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall**. The input values were successfully processed by the **Random Forest Classifier**, and the system accurately predicted the most suitable crop.

In addition to crop prediction, all implemented features were verified for proper functionality. The system correctly displayed the **prediction confi-**

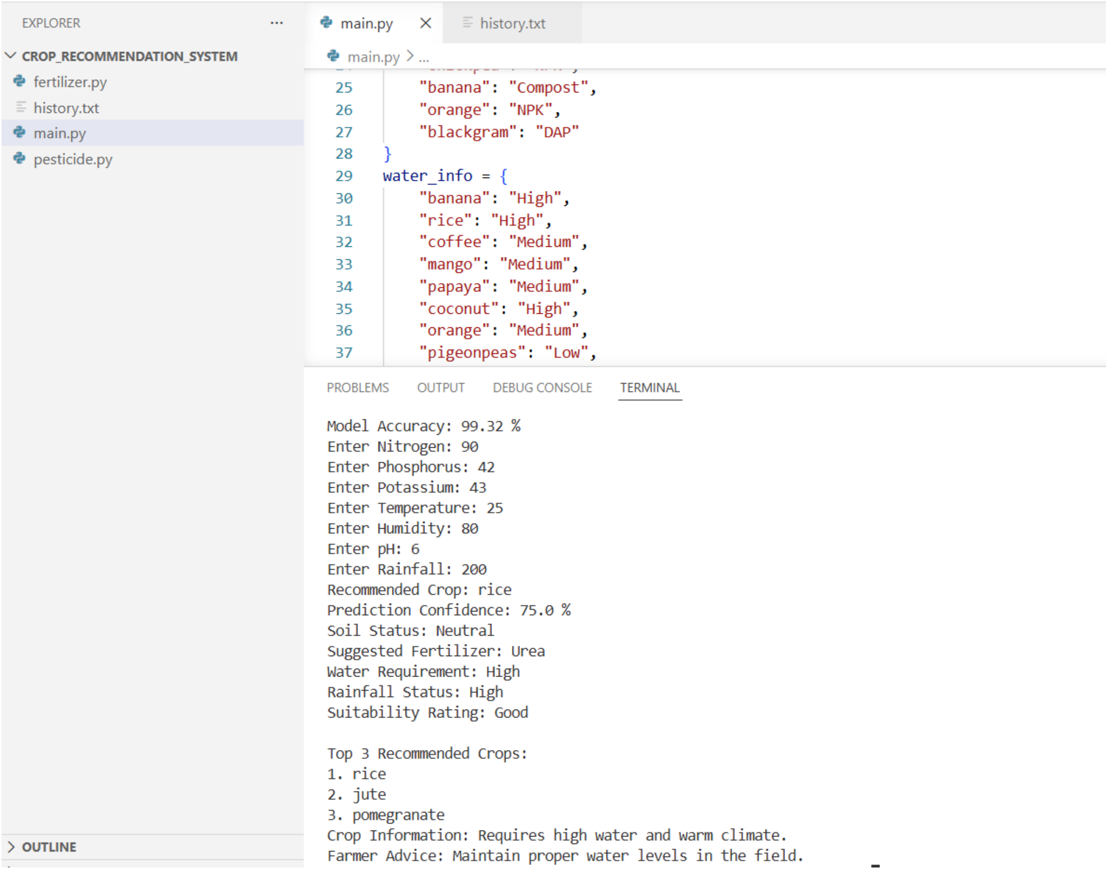

Figure 5.1: Simulation Results of the Crop Recommendation System

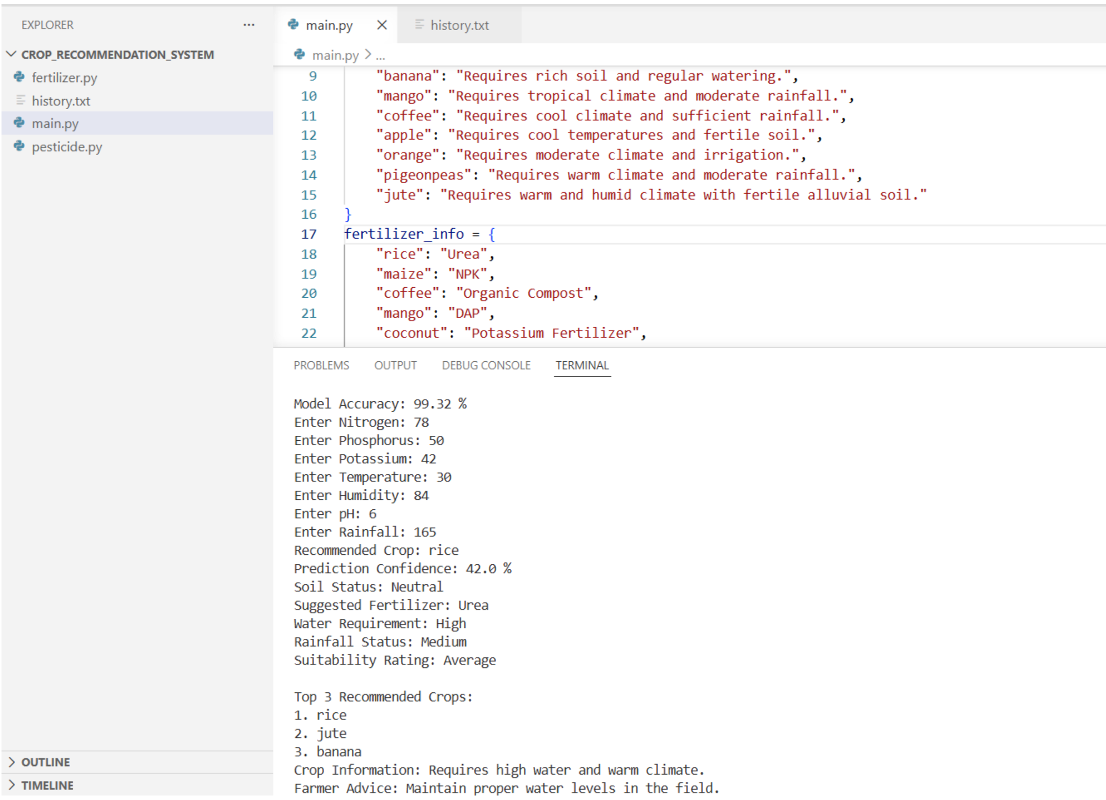

Figure 5.2: Simulation Results of the Crop Recommendation System

dence, **fertilizer suggestion**, **soil status (Acidic, Neutral, or Alkaline)**, **water requirement**, **rainfall status**, **suitability rating**, **top three recommended crops**, **crop information**, and **farmer advice**. The prediction history was also stored successfully for future reference. The obtained outputs matched the expected results, confirming the correctness and reliability of the implemented system. The functional verification demonstrates that the proposed application performs all intended operations accurately and can effectively assist users in making informed crop selection decisions.

5.4 Discussion

The results obtained from the proposed **Crop Recommendation System Using Machine Learning** demonstrate that the **Random Forest Classifier** is highly effective in predicting suitable crops based on soil and environmental parameters. The model achieved an accuracy of **99.32%**, indicating its ability to provide reliable and accurate crop recommendations. The use of important agricultural features such as **Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall** contributed significantly to the high prediction performance.

In addition to crop recommendation, the system provides several value-added features, including **prediction confidence, fertilizer suggestion, soil status analysis, water requirement, rainfall status, suitability rating, top three recommended crops, crop information, and farmer advice**. These features enhance the practical usefulness of the system by providing farmers with comprehensive information for better agricultural decision-making.

The developed system is simple, efficient, and easy to use, requiring only basic agricultural input parameters to generate meaningful recommendations. Although the model performs well on the available dataset, its performance may vary under different environmental conditions or with unseen agricultural data. Future improvements such as integrating **real-time weather data**, expanding the dataset, and deploying the system as a **web or mobile application** can further improve its accuracy, usability, and real-world applicability.

5.5 Summary

This chapter presented the implementation, functional verification, and performance evaluation of the proposed **Crop Recommendation System Using Machine Learning**. The system was successfully developed using **Python** and the **Random Forest Classifier**, achieving a prediction accuracy of **99.32%**. The model effectively analyzed soil nutrients and environmental parameters to recommend the most suitable crop for cultivation.

The chapter also demonstrated the successful implementation of additional features, including **prediction confidence, fertilizer suggestion, soil status analysis, water requirement, rainfall status, suitability rating, top three crop recommendations, crop information, and farmer advice**. The obtained results confirmed the reliability and effectiveness of the proposed system in supporting agricultural decision-making. Overall, the developed application provides an efficient and user-friendly solution for crop recommendation and serves as a foundation for future enhancements such as real-time weather integration and web or mobile application deployment.

CHAPTER 6

Conclusions and Future Scope

6.1 Introduction

The chapter summarizes the overall outcomes of the proposed **Crop Recommendation System Using Machine Learning** and highlights its significance in supporting smart agricultural practices. The project successfully utilizes the **Random Forest Classifier** to recommend the most suitable crop based on soil nutrients and environmental parameters, including **Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall**. The developed system achieved a prediction accuracy of **99.32%**, demonstrating its effectiveness and reliability in crop recommendation.

In addition to crop prediction, the system provides several useful features such as **prediction confidence, fertilizer suggestion, soil status analysis, water requirement, rainfall status, suitability rating, top three crop recommendations, crop information, and farmer advice**, making it a comprehensive decision-support tool for farmers. This chapter concludes the project by summarizing its achievements and discussing possible future enhancements, including **real-time weather integration, larger datasets, and deployment as a web or mobile application** to improve accessibility and practical usability.

6.2 Summary of Work

The proposed **Crop Recommendation System Using Machine Learning** was successfully designed, developed, and implemented to assist farmers in selecting the most suitable crop based on soil nutrients and environmental conditions. The system uses the **Random Forest Classifier** to analyze key agricultural parameters such as **Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall**, and recommends the most appropriate crop for cultivation. The model achieved a high prediction accuracy of **99.32%**, demonstrating its effectiveness and reliability.

In addition to crop recommendation, the system includes several enhanced features such as **prediction confidence, fertilizer suggestion, soil status analysis, water requirement, rainfall status, suitability rating, top three recommended crops, crop information, and farmer advice**. These features provide comprehensive support to users and improve agricultural decision-making. The project was implemented using **Python**, with **Pandas, NumPy, Scikit-learn, and Matplotlib** libraries for data processing, model training, evaluation, and visualization. Overall, the developed system provides an efficient, user-friendly, and reliable solution for smart crop recommendation and contributes towards promoting sustainable agricultural practices.

6.3 Key Observations

The implementation and evaluation of the proposed **Crop Recommendation System Using Machine Learning** led to several important observations. The **Random Forest Classifier** achieved a high prediction accuracy of **99.32%**, indicating that it is highly effective for crop recommendation using soil and environmental parameters. The input features, namely **Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall**, played a significant role in improving the prediction performance.

The additional functionalities, including **prediction confidence, fertilizer suggestion, soil status analysis, water requirement, rainfall status, suitability rating, top three recommended crops, crop information, and farmer advice**, enhanced the usability of the system and provided valuable information for better agricultural decision-making. The system demonstrated fast prediction with minimal computational resources and was easy to use. These observations indicate that the proposed solution is reliable, efficient, and suitable for supporting smart and sustainable farming practices.

6.4 Conclusion

The proposed **Crop Recommendation System Using Machine Learning** was successfully designed and implemented to recommend the most suitable crop based on soil nutrients and environmental conditions. The system utilizes the **Random Forest Classifier** to analyze important agricultural parameters, including **Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall**, and provides accurate crop recommendations. The model achieved a high prediction accuracy of **99.32%**, demonstrating its effectiveness and reliability in crop prediction.

In addition to crop recommendation, the system provides several useful features such as **prediction confidence, fertilizer suggestion, soil status analysis, water requirement, rainfall status, suitability rating, top three recommended crops, crop information, and farmer advice**. These features enhance the usability of the application and support farmers in making informed agricultural decisions. Overall, the developed system is efficient, user-friendly, and contributes to **smart and sustainable agriculture** by helping improve crop selection and resource utilization.

The developed system was also capable of accepting new patient information as input and predicting whether the patient is likely to be diabetic or non-diabetic. In addition, graphical visualizations such as the diabetes distribution graph and algorithm accuracy comparison chart were generated to provide better insight into the dataset and model performance. These visualizations help in understanding the data distribution and evaluating the effectiveness of the implemented algorithms.

Although the developed model does not replace professional medical diagnosis, it serves as an efficient decision-support tool for healthcare professionals by assisting in early diabetes screening. The project demonstrates how machine learning can analyze medical data, identify hidden patterns, and support informed clinical decisions. Overall, the objectives of the project were successfully achieved, and the developed Diabetes Prediction System provides a simple, efficient, and practical solution for preliminary diabetes prediction.

This work also establishes a strong foundation for future improvements using larger datasets, advanced machine learning algorithms, and real-time healthcare applications.

With further optimization and hardware implementation, the proposed R4LC architecture has the potential to become an efficient computational unit for real-time signal processing systems.

CHAPTER 7

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

7.1 Contribution to Program Outcomes (POs)

The proposed project titled **"Crop Recommendation System Using Machine Learning"** contributes to the attainment of several Program Outcomes defined for the Electronics and Communication Engineering program. The project applies programming skills, data preprocessing techniques, and machine learning algorithms to develop an intelligent crop recommendation system. It analyzes soil and environmental parameters to recommend the most suitable crop and provides additional information such as fertilizer suggestions, water requirements, soil status, crop information, and farmer advice. The project demonstrates the practical application of artificial intelligence and data-driven decision-making in smart agriculture.

PO1: Engineering Knowledge

The project applies engineering knowledge in programming, machine learning, and data analysis to develop an intelligent crop recommendation system. Python programming and the Random Forest algorithm are used to analyze agricultural data and generate accurate crop recommendations.

PO2: Problem Analysis

The project addresses the problem of selecting suitable crops based on soil nutrients and climatic conditions. It analyzes parameters such as Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall to recommend the most appropriate crop.

PO3: Design and Development of Solutions

An intelligent machine learning-based crop recommendation system is designed and implemented using the Random Forest Classifier. The system predicts suitable crops and provides additional recommendations including fertilizer suggestions, soil status, water requirements, crop information, and farmer advice.

PO4: Investigation of Complex Problems

The developed model is trained and tested using agricultural datasets. Model performance is evaluated using prediction accuracy, and the Random Forest algorithm achieved an accuracy of **99.32%**, demonstrating its effectiveness for crop prediction.

PO5: Modern Tool Usage

The project utilizes modern software tools including Python, Pandas, NumPy, Scikit-learn, Matplotlib, and Visual Studio Code for data preprocessing, model development, prediction, and performance evaluation.

PO6: Engineer and Society

The developed crop recommendation system supports farmers by providing scientific recommendations for crop selection. Better crop planning improves agricultural productivity, reduces resource wastage, and contributes to sustainable farming practices.

7.2 Program Specific Outcomes (PSOs)

PSO1: Ability to develop software solutions using modern programming tools

The project demonstrates the ability to develop an intelligent software application using Python and machine learning libraries. It includes data preprocessing, model training, prediction, and result analysis using modern software development tools.

PSO2: Application of computational techniques for solving real-world problems

The project applies machine learning techniques to solve a real-world agricultural problem. It assists farmers in selecting suitable crops based on environmental conditions, thereby improving productivity and supporting precision agriculture.

7.3 Alignment with Sustainable Development Goals (SDGs)

The proposed project contributes to several United Nations Sustainable Development Goals by promoting smart agriculture through machine learning and data-driven decision-making.

SDG 2: Zero Hunger

The project supports sustainable agriculture by recommending suitable crops based on soil and environmental conditions, thereby improving crop productivity and food security.

SDG 9: Industry, Innovation and Infrastructure

The project promotes technological innovation by applying machine learning algorithms in agriculture to support intelligent farming solutions.

SDG 12: Responsible Consumption and Production

The system encourages efficient utilization of soil nutrients, fertilizers, and water resources, reducing wastage and promoting sustainable agricultural practices.

SDG 13: Climate Action

The project considers environmental parameters such as temperature, humidity, and rainfall while recommending crops, helping farmers adapt to changing climatic conditions and encouraging climate-resilient agriculture.

7.4 Overall Impact

The proposed **Crop Recommendation System Using Machine Learning** has a positive impact on modern agriculture by assisting farmers in selecting the most suitable crop based on soil nutrients and environmental conditions. By utilizing the **Random Forest Classifier**, the system provides accurate crop recommendations with a prediction accuracy of **99.32%**, enabling better agricultural planning and decision-making. The additional features, including fertilizer suggestions, soil status analysis, water requirement estimation, rainfall status, suitability rating, top three crop recommendations, crop information, and farmer advice, further enhance the usefulness of the system.

The project promotes efficient utilization of agricultural resources, reduces the risk of unsuitable crop selection, and encourages sustainable farming practices. It also demonstrates the practical application of machine learning in solving real-world agricultural challenges. Overall, the developed system contributes to improved crop productivity, better resource management, and supports the adoption of smart farming technologies for sustainable agricultural development.